How much of breast cancer grade is in the routine record? A leakage-audited, multi-cohort open-data prediction study with MRI radiomics and transport to Saudi case-mix

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Abstract

Background: Nottingham grade guides breast cancer management but is a tissue-based, moderately reproducible judgement. How much of it routine variables carry, whether such models transport, and what they imply for Saudi patients is unknown. **Methods:** We harmonised 5025 grade-labelled invasive cancers from 7 open cohorts. Seven models, from a prevalence baseline to gradient boosting, were developed in METABRIC (2386 patients) with nested cross-validation, compared with logistic regression by paired bootstrap, and validated, frozen, in five external cohorts and by leave-one-dataset-out. We report discrimination, calibration, decision curves, a leakage audit, an MRI-radiomics extension and re-weighting to published Saudi case-mix (TRIPOD+AI). **Results:** Internal macro-AUC was 0.72–0.73 for every learned model; XGBoost minus logistic regression was +0.001 (95% CI -0.006 to +0.008). External macro-AUCs were 0.69–0.71 (MSK-2018) and 0.70–0.72 (Duke). Calibration slopes stayed near one, but grade-3 calibration-in-the-large failed in opposite directions (intercepts +0.99 and -1.02) and was corrected by an intercept update. Leaky protocols inflated macro-AUC by up to +0.297. Adding 529 DCE-MRI radiomic features did not help (Δ -0.045, -0.077 to -0.011). Saudi case-mix raised each cohort's grade-3 share by +9.6 to +10.3 points, yet a METABRIC-developed model would predict 59.1% grade 3 against 36.7% published. **Conclusions:** Routine variables carry moderate, transportable information about grade; logistic regression suffices and calibration must be updated locally. Western-developed models would over-predict high-grade disease in Saudi patients.

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1. Introduction

Grade matters, but it is assigned late and inconsistently. Nottingham (Elston–Ellis) histological grade combines tubule formation, nuclear pleomorphism and mitotic count into a three-level score that remains one of the strongest independent prognostic factors in invasive breast cancer [1–3]. It is, however, a tissue-based judgement made after biopsy or excision, and it is only moderately reproducible: agreement between departments and pathologists is typically in the κ range 0.5–0.7 [4, 5], grading proportions vary significantly between and within laboratories in national data [6], and the intermediate grade 2, which accounts for a large share of tumours, is prognostically heterogeneous [7, 8]. These facts motivate a simple but rarely answered question: *how much of a tumour's grade is already carried by the routinely recorded clinicopathological variables*—age, menopausal status, ER, PR and HER2 status, histological type, tumour size and nodal status? A credible answer would serve as a baseline against which image-based grade predictors are judged, as a tool for imputing grade where registry records lack it, and as a quality-assurance signal when an assigned grade is improbable given the rest of the record.

Computational grading has moved to images without a tabular baseline. Recent work predicts grade from whole-slide images with deep learning and pathology foundation models [9–11] and from dynamic contrast-enhanced MRI radiomics [12, 13]. We found no study that quantified, on open multi-cohort data, how well the routine clinicopathological record alone predicts three-level grade, how such a model behaves outside its development population, or how much of the reported performance in this area could be produced by analytic leakage. Leakage—information about the outcome reaching the model through predictors that encode it, through preprocessing fitted before data splitting, or through repeated records of the same patient—is a recognised cause of irreproducible machine-learning results in medicine [14–16]. Grade is unusually exposed to it, because widely shared datasets carry composite indices (the Nottingham Prognostic Index) and component scores that contain grade by construction. Separately, systematic reviews show that flexible learners seldom outperform regression for clinical prediction [17], that calibration is the most frequently neglected property of prediction models [18], and that external validation and reporting should follow TRIPOD+AI [19–21].

The Saudi and Gulf context, and the problem of transport. Breast cancer is the most common cancer among Saudi women; its age-standardised incidence in Saudi females rose from 11.8 to 29.7 per 100,000 between 2001 and 2017, when it accounted for 30.9% of female cancers, with a median age at diagnosis of about 51 years [22]. Patients in Arab countries present younger and more often with hormone-receptor-negative and HER2-positive disease than patients in Western screening cohorts [23–27]. Yet no grade-labelled patient-level dataset from the region is openly available, and local analyses are typically based on single-centre hospital records that cannot be shared or independently checked [27, 28]. A model developed elsewhere can only be used in a new population if it *transports*: case-mix differences can change both its discrimination and, above all, its calibration [29–31]. Whether Western grade models transport to Saudi case-mix is therefore an open and practically important question that can be approached, without patient data, by re-weighting open cohorts to published Saudi aggregates.

Design principles. Clinicopathological analyses of grade are prone to avoidable errors: correlation coefficients computed on arbitrary codes of nominal categories, causal readings of cross-sectional associations, in-situ carcinoma counted as invasive disease, lobular carcinoma merged with ductal carcinoma, and validation designs in which imputation, resampling or repeated records let information cross from test to training data [14, 15]. We therefore fixed the harmonisation rules, feature sets, models, metrics and expected directions of effect before running the experiments, and used only openly downloadable cohorts, so that every number in this paper can be regenerated by a reviewer.

Contributions.

1. An open, checksummed, harmonised common data model of 5025 grade-labelled invasive breast cancers from 7 cohorts in three continents, with explicit rules that prevent the in-situ, histology and duplication errors described above (Figure 1).
2. A TRIPOD+AI-compliant model ladder from a prevalence baseline to gradient boosting, with nested cross-validation, multiple imputation, paired comparisons against logistic regression, external validation in two full-feature and three core-feature cohorts, leave-one-dataset-out validation, calibration, model updating and decision curves.
3. A leakage audit that quantifies, on the same data and models, how much four common shortcuts inflate apparent discrimination.
4. An imaging extension in the Duke breast MRI cohort that asks whether 529 DCE-MRI radiomic features add to the routine clinical record.
5. A case-mix transport analysis that re-weights open cohorts to published Saudi

hospital series and shows what Western-developed grade models would imply for Saudi patients.

2. Methods

2.1. Design, reporting and framework

This is a secondary analysis of de-identified, publicly released data. The prediction component is reported according to TRIPOD+AI [19] and the descriptive component according to STROBE [32]; the full TRIPOD+AI checklist is in Supplementary Table S9. Figure 1 summarises the framework: open sources are harmonised to a common data model (lane A); the model is developed and validated (lane B); and the results are stress-tested by a leakage audit, a plausibility check, an imaging extension and a transport analysis (lane C). Feature sets, model ladder, metrics, expected directions of effect and raking targets were fixed in `configs/experiment.yaml` before the experiments were run (seed 20261017). Every table, figure and in-text number is produced by code from the results files; none is typed by hand.

2.2. Data sources

Tabular data were obtained without registration from the cBioPortal data hub [33, 34], The Cancer Imaging Archive and the UCI Machine Learning Repository. A download script records SHA-256 checksums so that silent upstream changes are detectable. Cohorts and roles were:

- **METABRIC** (UK/Canada; development) [35, 36]: surgical cohort with Nottingham grade, tumour size (mm), positive-node count, receptors and inferred menopausal state.
- **MSK-2018** (USA; external, full feature set) [37]: patients sequenced at Memorial Sloan Kettering, enriched for advanced hormone-receptor-positive disease, with diagnosis-time grade, T and N category, receptors and menopausal status.
- **Duke-Breast-Cancer-MRI** (USA; external, full feature set; imaging extension) [38]: 922 patients with invasive cancer imaged at Duke University Hospital (2000–2014), with Nottingham grade and its component scores, receptors, T and N category, menopausal status and 529 pre-computed DCE-MRI features (CC BY-NC 4.0; files obtained from a checksummed public mirror of the TCIA release).
- **MBCproject** (USA/Canada; external, core features) [39, 40]: patient-partnered metastatic cohort with abstracted diagnosis-time records; age in five-year bands.
- **Sanger** (UK/Europe; external, core) [41]: 100 primary tumours with grade and the three Nottingham component scores.

- **Broad** (Mexico and Vietnam; external, core) [42]: the only open grade-labelled cohort from low- and middle-income countries we identified.
- **BCCRC** (Canada; triple-negative; descriptive) [43] and **TCGA-BRCA** (USA; descriptive, no structured grade) [44].
- **Ljubljana** (Slovenia; sensitivity) [45]: 286 patients with "degree of malignancy" 1–3 and binned covariates.

No openly downloadable, grade-labelled patient-level cohort from Saudi Arabia or the Gulf was identified; Saudi relevance enters through published aggregates (Section 2.12).

2.3. Harmonisation to a common data model

Each source was mapped to one row per patient: age (years), post-menopausal status, ER, PR and HER2 status (positive/negative; equivocal or borderline treated as missing), histological type (no-special-type/ductal, lobular, mixed ductal–lobular, other special type), T category (T1 ≤20 mm, T2 21–50 mm, ≥T3 including T4) and nodal status (any positive regional node, including micrometastasis, versus none; isolated tumour cells node-negative). Four rules corrected the errors described in the Introduction: (i) the outcome is invasive Nottingham or differentiation grade 1–3, anything else missing; (ii) pure in-situ disease (DCIS, LCIS, METABRIC stage 0) and non-carcinomas are excluded and never relabelled; (iii) lobular carcinoma is never merged with ductal carcinoma; (iv) men are excluded. Where several sequenced samples existed per patient (MSK-2018) the diagnosis-time record was used once. Participant flow is in Supplementary Table S1.

2.4. Outcome, predictors and feature sets

The primary outcome was three-class grade; the key secondary outcome was grade 3 versus 1–2. The *full* feature set (age, menopausal status, ER, PR, HER2, histology, T category, N status) was used where available (METABRIC, MSK-2018, Duke); the *core* set (age, ER, PR, HER2, histology) is shared by all external cohorts. The Nottingham Prognostic Index, the Nottingham component scores and stage were quarantined: they were never predictors and were used only in the leakage audit; a unit test fails the build if any enters a feature set.

2.5. Descriptive and associational statistics

Association measures were matched to measurement scale: Kendall's τ_b for grade against binary or ordinal variables; bias-corrected Cramér's V [46] for nominal histology; Spearman's ρ with the Kruskal–Wallis ε^2 for age. All effects carry percentile-bootstrap 95% confidence intervals (2000 resamples) and p values were adjusted across the 22 tests with the Benjamini–Hochberg

procedure [47]. Associations are interpreted as cross-sectional co-occurrence, not causation. Multivariable associations are summarised by a proportional-odds model fitted to complete cases in METABRIC with bootstrap intervals.

2.6. Model development

Seven models formed a ladder of increasing flexibility: a prevalence (majority-class) baseline, multinomial logistic regression (LR) with an L_2 penalty, proportional-odds logistic regression, k-nearest neighbours, random forest, XGBoost [48] and LightGBM [49]. A tabular foundation model, TabPFN [50], was pre-specified as an optional rung (status: not run; see Limitations). Every learned step—chained-equation imputation [51], standardisation and the classifier—was wrapped in one pipeline and re-fitted inside each training fold. Hyper-parameters were tuned by grid search minimising log-loss in an inner 3-fold cross-validation. Class imbalance was not corrected by resampling, because resampling distorts predicted probabilities [52]; hard class assignments used the Bayes rule under equal misclassification costs, $\hat{y} = \arg\max_k p_k/\pi_k$, with π_k the training-fold prevalence.

Missing data. The primary analysis used single chained-equation imputation fitted within folds. As pre-specified sensitivity analyses we used (a) multiple imputation with $m = 20$ draws from the posterior predictive distribution, a separate model per imputed dataset and averaging of the predicted probabilities ("pooled predictions"), which is the recommended approach when missing values are expected at deployment [53, 54]; and (b) a complete-case analysis. Both used the same nested resampling as the primary analysis.

Sample size. The minimum sample size for developing the multinomial model was calculated with the Riley criteria [55] applied to each pair of outcome categories as proposed for multinomial outcomes [56]: global shrinkage ≥0.9 and a small optimism in apparent R^2 (δ=0.05) for every pairwise sub-model with 10 predictor parameters, and estimation of each grade prevalence within ±0.05. The anticipated pairwise Cox–Snell R^2 values were taken from out-of-fold LR predictions, which are free of optimism.

2.7. Validation

Internal validation used nested, 10-times repeated, stratified 5-fold cross-validation in METABRIC; out-of-fold probabilities were averaged across repeats and patient-level bootstrap 95% CIs (2000 resamples) were computed on these averages. Each model was compared with LR by a paired bootstrap of the difference in macro-AUC on the same resamples. For external validation each model was re-tuned on all of METABRIC, frozen, and applied unchanged to MSK-2018 and Duke (full set) and to MSK-2018, Duke, MBCproject, Sanger and Broad (core set).

Internal–external validation used leave-one-dataset-out (LODO) training across METABRIC, MSK-2018, Duke, MBCproject and Sanger with the core set [57].

2.8. Performance measures

Discrimination was summarised by the macro-averaged one-versus-rest AUC and per-grade AUCs; classification by macro-F1, balanced accuracy and quadratic-weighted κ (QWK); overall probabilistic accuracy by the multiclass Brier score. For grade 3 versus 1–2 we report AUC, calibration intercept (calibration-in-the-large) and slope, LOWESS calibration curves [18] and decision curves [58]. Multiclass calibration-in-the-large is summarised as observed versus mean predicted prevalence of every grade [59]. In each external cohort an intercept-only update, cross-fitted over two folds, separated miscalibration caused by a different grade prevalence from miscalibration of the predictor effects [21].

2.9. Leakage audit

Four shortcuts were quantified against the correct protocol on the same patients and models [14]: (a) the Nottingham Prognostic Index, which contains grade, as a predictor; (b) imputation and SMOTE [60] applied to the whole dataset before cross-validation; (c) the Nottingham component scores as predictors (Sanger and Duke); and (d) splitting by record instead of by patient when patients contribute several records (MSK-2018), the tabular analogue of tile-level splitting in histopathology [61].

2.10. Interpretability and clinical plausibility

For XGBoost we computed TreeSHAP values for the grade-3 output [62, 63]. Before the analysis we specified the expected direction of association with grade 3 for seven features (ER−, PR−, HER2+, larger T, node-positive, younger age and non-lobular histology associated with higher grade). Agreement was recorded when the bootstrap interval of the Spearman correlation between a feature's value and its SHAP value excluded zero in the expected direction.

2.11. Imaging extension: DCE-MRI radiomics in Duke

Within the graded Duke patients we compared three L_2-penalised multinomial LR models by nested 10×5-fold cross-validation: the clinical full set; 529 pre-computed DCE-MRI radiomic features [38] (median-imputed and standardised in-fold); and their fusion. The penalty was tuned in an inner 3-fold loop. The added value of radiomics was the paired-bootstrap difference in macro-AUC versus the clinical model.

2.12. Transport to Saudi case-mix

We re-weighted the METABRIC, MSK-2018 and Duke analysis sets by iterative proportional fitting (raking) to the published marginal distributions of Saudi hospital series and asked what grade distribution and model predictions the Saudi case-mix implies [30]. The primary anchor was a series of Saudi nationals from King Fahd University Hospital, Dammam [27] (ER, PR, HER2, histology; mixed tumours counted with ductal; share aged ≤50 years assumed 50% from the reported median of 51, varied from 40% to 60% in sensitivity analysis). A second anchor was a Riyadh tertiary-centre series [28] (ER, PR, HER2, histology, share aged ≤50 years; receptor proportions recomputed from the published counts). All anchor values were transcribed from the published tables, with the table and page recorded in `configs/anchors.yaml`. For each anchor and cohort we report the re-weighted grade distribution with a bootstrap 95% CI (patients resampled and re-raked 500 times), the effective sample size (ESS) of the weights, the weighted mean predicted grade probabilities, and weighted discrimination. Published Saudi grade-3 proportions from all series reporting them were pooled with a DerSimonian–Laird random-effects model on the logit scale. This analysis shows how far measured case-mix differences explain grade differences; it cannot reproduce local grading practice or unmeasured differences.

2.13. Software, reproducibility and ethics

Analyses used Python 3.13 with scikit-learn 1.9.1, imbalanced-learn 0.14.2, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1 and statsmodels 0.15.0 (versions pinned in the repository). One command (`make all`) downloads and checksums the data, runs the unit tests that guard the harmonisation and leakage rules, executes stages E1–E10 and regenerates every table, figure and in-text number. The data are de-identified and publicly released under open terms; no new data were collected, and no ethics approval or consent was required for this secondary analysis.

3. Results

3.1. Cohorts

After exclusions, 5025 graded patients in 7 cohorts and 1072 ungraded TCGA patients were analysed (Table 1; flow in Supplementary Table S1). Case-mix differed widely (Figure 2). Median age ranged from 43 years (MBCproject) to 61 years (METABRIC); grade 3 accounted for 50.2% of METABRIC, 67.6% of MSK-2018 (enriched for advanced disease) and 32.5% of Duke. The Dammam Saudi series [27] had fewer ER-positive (57.3%) and PR-positive (45.0%) tumours than any open cohort except the triple-negative BCCRC series and more HER2-positive tumours (25.3%) than all but Sanger, yet a grade distribution (8.7/54.5/36.7%) weighted more to grade 2 than METABRIC.

Table 1: Characteristics of the open cohorts after harmonisation and of two published Saudi hospital series. Percentages are of patients with the variable recorded (missingness in `results/e1_cohorts.csv`). MSK = MSK-2018; MBCP = MBCproject; Broad = Mexico and Vietnam; dev/ext/desc = development, external validation, descriptive. †Saudi nationals, King Fahd University Hospital, Dammam, 1998–2017 [27]. ‡King Fahad Medical City, Riyadh, 2016–2019 (90.1% Saudi nationals; 39.8% stage IV) [28]; receptor percentages recomputed from the published counts.

Role	METABRIC dev	MSK ext	Duke ext	MBCP ext	Sanger ext	Broad ext	BCCRC desc	TCGA desc	Dammam† anchor	Riyadh‡ anchor
Patients analysed, *n*	2386	1573	637	207	100	59	63	1072	286	283
Age, median (IQR), y	61 (51–70)	52 (43–61)	52 (45–61)	43 (38–53)	55 (44–66)	52 (42–58)	56 (48–63)	58 (49–67)	51	–
Age ≤ 50 y, %	23.4	45.7	42.4	70.0	43.0	45.8	33.3	30.4	–	45.2
Grade 1, %	8.9	5.7	17.7	12.1	8.0	6.8	6.3	–	8.7	4.2
Grade 2, %	40.9	26.7	49.8	50.2	41.0	49.2	4.8	–	54.5	47.0
Grade 3, %	50.2	67.6	32.5	37.7	51.0	44.1	88.9	–	36.7	48.8
IDC / NST, %	76.7	81.8	89.0	83.0	79.2	96.6	92.1	77.2	91.3	–
ILC, %	7.9	11.7	9.9	10.2	6.2	0.0	0.0	20.2	7.0	–
Mixed ductal–lobular, %	11.2	4.9	0.0	5.8	9.4	1.7	1.6	0.0	–	–
Other special type, %	4.2	1.5	1.1	1.0	5.2	1.7	6.3	2.6	1.7	–
ER positive, %	73.9	84.6	75.5	84.1	79.0	68.6	4.8	–	57.3	62.6
PR positive, %	52.1	72.2	64.8	74.7	60.0	58.8	1.6	–	45.0	54.1
HER2 positive, %	12.5	11.5	17.6	23.6	30.0	17.4	0.0	–	25.3	36.1
T1 (≤20 mm), %	43.8	49.2	44.9	–	–	–	42.6	25.8	–	–
T2 (21–50 mm), %	51.0	34.4	42.7	–	–	–	49.2	57.8	–	–
≥T3, %	5.2	16.3	12.4	–	–	–	8.2	16.4	–	–
Node positive, %	47.6	50.3	40.3	–	–	–	49.1	51.7	–	–
Post-menopausal, %	78.0	48.0	53.8	28.1	–	30.8	–	–	–	–

3.2. Associations between grade and routine variables

Of 22 pre-specified associations, 20 remained significant after Benjamini–Hochberg adjustment (Table 2, Figure 3). In METABRIC, higher grade co-occurred most strongly with ER negativity (τ_b -0.37) and PR negativity (τ_b -0.31), followed by HER2 positivity (+0.19), larger T category (+0.15), nodal involvement (+0.14) and younger age (ρ -0.12); histological type had a moderate association (V = 0.18). Directions were the same in MSK-2018. In the multivariable proportional-odds model (1800 complete cases; Table 5), ER positivity (odds ratio for higher grade 0.17, 95% CI 0.12–0.24) and lobular histology (0.41, 0.28–0.57) remained the strongest independent correlates.

3.3. Internal validation: a flat model ladder

Every learned model discriminated grade well above the prevalence baseline, and the ladder was flat (Table 3, Figure 4). Multinomial LR (0.725) and the proportional-odds model (0.725) matched XGBoost (0.723) and the other ensembles: the paired difference XGBoost minus LR was +0.001 (95% CI -0.006 to +0.008), and no model's macro-AUC exceeded LR's by more than 0.009 at the upper confidence limit; k-NN was lowest (0.715). Variation across repeats was negligible (SD at most 0.003). Discrimination was best for grade 3 versus the rest (AUC 0.752 for Ordinal LR (prop. odds)) and weakest for grade 2. With the prevalence-adjusted decision rule, macro-F1 was 0.465 for LR against 0.223 for the baseline. Multiple imputation with pooled predictions changed macro-AUC by at most 0.003 (Supplementary

Table 2: Scale-appropriate associations with bootstrap 95% CIs and Benjamini–Hochberg-adjusted p values. Binary variables are coded 1 = present/positive, so a negative τ_b for ER means ER-positive tumours tend to be lower grade.

Pair	Measure	METABRIC		MSK-2018	
		Effect (95% CI)	p_{BH}	Effect (95% CI)	p_{BH}
Grade × ER+	τ_b	-0.373 (-0.404, -0.342)	<0.001	-0.211 (-0.241, -0.178)	<0.001
Grade × PR+	τ_b	-0.310 (-0.350, -0.271)	<0.001	-0.203 (-0.245, -0.165)	<0.001
Grade × HER2+	τ_b	+0.194 (+0.155, +0.229)	<0.001	+0.143 (+0.106, +0.181)	<0.001
Grade × T category	τ_b	+0.153 (+0.115, +0.191)	<0.001	+0.210 (+0.166, +0.257)	<0.001
Grade × Node+	τ_b	+0.137 (+0.097, +0.178)	<0.001	+0.172 (+0.117, +0.221)	<0.001
Grade × Post-menopausal	τ_b	-0.104 (-0.148, -0.061)	<0.001	-0.043 (-0.091, +0.004)	0.083
Age × Grade	ρ	-0.118 (-0.158, -0.078); ε^2=0.016	<0.001	-0.123 (-0.171, -0.074); ε^2=0.016	<0.001
Grade × Histology	V	+0.177 (+0.145, +0.217)	<0.001	+0.108 (+0.076, +0.157)	<0.001
Histology × ER+	V	+0.139 (+0.109, +0.174)	<0.001	+0.205 (+0.158, +0.263)	<0.001
T category × Node+	τ_b	+0.218 (+0.179, +0.256)	<0.001	+0.374 (+0.330, +0.417)	<0.001
Age × ER+	ρ	+0.267 (+0.227, +0.304); ε^2=0.071	<0.001	+0.017 (-0.033, +0.065); ε^2=0.000	0.488

Table S5); a complete-case analysis (1800 patients) gave 0.729 for LR. The Riley sample-size criteria were not met (4477 patients required, 2386 available); the binding constraint was the grade 1 versus grade 2 sub-model, whose anticipated R^2 is small (Supplementary Table S6).

Table 3: Internal validation in METABRIC (nested, 10× repeated stratified 5-fold cross-validation; full feature set). Point estimates are means over repeats; 95% CIs are patient-level bootstraps (2000 resamples) of repeat-averaged out-of-fold predictions. Δ is the paired-bootstrap difference in macro-AUC versus multinomial LR. QWK = quadratic-weighted κ.

Model	Macro-AUC (95% CI)	Δ vs LR (95% CI)	AUC G1	AUC G2	AUC G3	Macro-F1 (95% CI)	QWK	Brier
Majority class	0.499 (0.478–0.514)	-0.228 (-0.255, -0.203)	0.497	0.500	0.499	0.223 (0.217–0.229)	0.000	0.573
Multinomial LR	0.725 (0.707–0.742)	ref.	0.741	0.681	0.752	0.465 (0.445–0.486)	0.356	0.490
Ordinal LR (prop. odds)	0.725 (0.709–0.743)	+0.001 (-0.004, +0.006)	0.744	0.679	0.752	0.458 (0.435–0.476)	0.366	0.491
k-NN	0.715 (0.699–0.735)	-0.008 (-0.016, +0.001)	0.729	0.676	0.741	0.452 (0.427–0.468)	0.315	0.496
Random forest	0.724 (0.708–0.743)	+0.001 (-0.007, +0.009)	0.739	0.682	0.751	0.449 (0.424–0.463)	0.344	0.492
XGBoost	0.723 (0.708–0.743)	+0.001 (-0.006, +0.008)	0.737	0.678	0.753	0.458 (0.440–0.480)	0.344	0.492
LightGBM	0.719 (0.705–0.741)	-0.001 (-0.010, +0.007)	0.728	0.678	0.752	0.457 (0.443–0.484)	0.338	0.496

3.4. External and leave-one-dataset-out validation

Applied unchanged with the full feature set, models reached macro-AUCs of 0.69–0.71 in MSK-2018 (LR 0.710, 95% CI 0.685–0.734) and 0.70–0.72 in Duke (LR 0.709, 0.681–0.737; 637 patients) (Table 4). With the core set shared by all cohorts, MSK-2018 discrimination fell to 0.64–0.67, while the smaller cohorts ranged up to 0.797, with wide intervals. Leave-one-dataset-out estimates ranged 0.659–0.778 (Supplementary Table S2). Of 28 paired external comparisons of a flexible model with LR, 1 favoured the flexible model and 1 favoured LR at the 95% level.

3.5. Calibration, updating and clinical utility

Internally, predictions were well calibrated (slope 1.00 for LR, 1.02 for XGBoost; Supplementary Table S3). Externally the slope stayed near one (MSK-2018 1.03;

Table 4: External validation of models developed in METABRIC and applied without refitting. Calibration slope is for P(grade 3).

Target (features)	Model	n	Macro-AUC (95% CI)	G3-AUC (95% CI)	Macro-F1	Cal. slope (G3)
MSK-2018 (full)	Majority class	1573	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.269	–
MSK-2018 (full)	Multinomial LR	1573	0.710 (0.685–0.734)	0.727 (0.701–0.753)	0.372	1.03
MSK-2018 (full)	Ordinal LR (prop. odds)	1573	0.713 (0.688–0.737)	0.725 (0.699–0.750)	0.370	1.02
MSK-2018 (full)	Random forest	1573	0.704 (0.678–0.728)	0.725 (0.699–0.751)	0.350	1.11
MSK-2018 (full)	XGBoost	1573	0.711 (0.686–0.735)	0.726 (0.701–0.753)	0.373	1.03
MSK-2018 (full)	LightGBM	1573	0.705 (0.681–0.730)	0.720 (0.693–0.746)	0.388	0.82
Duke MRI (full)	Majority class	637	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.164	–
Duke MRI (full)	Multinomial LR	637	0.709 (0.681–0.737)	0.779 (0.739–0.816)	0.494	0.96
Duke MRI (full)	Ordinal LR (prop. odds)	637	0.705 (0.678–0.731)	0.778 (0.738–0.814)	0.487	0.92
Duke MRI (full)	Random forest	637	0.717 (0.689–0.744)	0.789 (0.752–0.826)	0.467	1.01
Duke MRI (full)	XGBoost	637	0.720 (0.691–0.748)	0.789 (0.750–0.826)	0.486	0.97
Duke MRI (full)	LightGBM	637	0.720 (0.691–0.748)	0.786 (0.749–0.824)	0.484	0.88
MSK-2018 (core)	Majority class	1573	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.269	–
MSK-2018 (core)	Multinomial LR	1573	0.667 (0.642–0.692)	0.685 (0.658–0.711)	0.349	0.90
MSK-2018 (core)	Ordinal LR (prop. odds)	1573	0.670 (0.645–0.695)	0.682 (0.655–0.709)	0.357	0.89
MSK-2018 (core)	Random forest	1573	0.665 (0.639–0.690)	0.690 (0.663–0.717)	0.357	0.98
MSK-2018 (core)	XGBoost	1573	0.658 (0.632–0.684)	0.680 (0.652–0.707)	0.357	0.88
MSK-2018 (core)	LightGBM	1573	0.660 (0.633–0.686)	0.681 (0.654–0.708)	0.357	0.72
MBCproject (core)	Majority class	207	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.182	–
MBCproject (core)	Multinomial LR	207	0.701 (0.647–0.754)	0.771 (0.703–0.832)	0.486	1.16
MBCproject (core)	Ordinal LR (prop. odds)	207	0.715 (0.665–0.761)	0.762 (0.692–0.829)	0.513	1.08
MBCproject (core)	Random forest	207	0.693 (0.640–0.744)	0.753 (0.686–0.819)	0.440	1.12
MBCproject (core)	XGBoost	207	0.697 (0.642–0.750)	0.758 (0.689–0.822)	0.455	1.01
MBCproject (core)	LightGBM	207	0.703 (0.649–0.757)	0.760 (0.693–0.826)	0.485	0.93
Sanger (core)	Majority class	100	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.225	–
Sanger (core)	Multinomial LR	100	0.797 (0.712–0.872)	0.806 (0.714–0.882)	0.570	1.33
Sanger (core)	Ordinal LR (prop. odds)	100	0.763 (0.679–0.842)	0.801 (0.704–0.882)	0.495	1.25
Sanger (core)	Random forest	100	0.764 (0.680–0.840)	0.772 (0.676–0.856)	0.441	1.27
Sanger (core)	XGBoost	100	0.766 (0.678–0.847)	0.774 (0.677–0.859)	0.479	1.20
Sanger (core)	LightGBM	100	0.767 (0.684–0.845)	0.774 (0.675–0.862)	0.486	1.03
Broad (MX/VN) (core)	Majority class	59	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.204	–
Broad (MX/VN) (core)	Multinomial LR	59	0.784 (0.691–0.869)	0.737 (0.599–0.863)	0.663	0.90
Broad (MX/VN) (core)	Ordinal LR (prop. odds)	59	0.777 (0.687–0.863)	0.738 (0.604–0.866)	0.675	0.86
Broad (MX/VN) (core)	Random forest	59	0.795 (0.678–0.891)	0.793 (0.653–0.910)	0.403	1.30
Broad (MX/VN) (core)	XGBoost	59	0.787 (0.668–0.882)	0.773 (0.634–0.888)	0.434	1.04
Broad (MX/VN) (core)	LightGBM	59	0.785 (0.665–0.887)	0.760 (0.616–0.884)	0.562	0.96
Duke MRI (core)	Majority class	637	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.164	–
Duke MRI (core)	Multinomial LR	637	0.702 (0.675–0.729)	0.783 (0.743–0.820)	0.434	1.01
Duke MRI (core)	Ordinal LR (prop. odds)	637	0.705 (0.678–0.732)	0.782 (0.743–0.819)	0.476	0.98
Duke MRI (core)	Random forest	637	0.715 (0.686–0.743)	0.782 (0.744–0.821)	0.485	1.02
Duke MRI (core)	XGBoost	637	0.714 (0.684–0.743)	0.784 (0.746–0.822)	0.476	0.97
Duke MRI (core)	LightGBM	637	0.712 (0.683–0.739)	0.779 (0.740–0.817)	0.487	0.88

Duke 0.96), so predictor effects transported, but calibration-in-the-large did not: in MSK-2018 observed grade 3 prevalence was 0.68 against a mean prediction of 0.47 (intercept +0.99), and in Duke 0.32 against 0.52 (intercept -1.02). A cross-fitted intercept update removed this miscalibration without affecting discrimination (Figure 5; per-grade observed and expected prevalences in Supplementary Table S4). Decision curves showed positive net benefit over treating all or no patients as grade 3 for thresholds from about 0.26 internally and 0.63 in MSK-2018 (where the high grade-3 prevalence makes "treat all" a strong comparator), and between 0.19 and 0.64 in Duke.

3.6. Interpretability and clinical plausibility

TreeSHAP ranked ER positive first and T category second (Figure 6). 7 of 7 pre-specified directions matched clinical expectation with bootstrap intervals excluding zero (Table 5) and agreed in sign with the proportional-odds odds ratios.

Table 5: Clinical-plausibility check. "Expected" was pre-specified. Ordinal odds ratios are for higher grade from a complete-case proportional-odds model (bootstrap 95% CI).

Feature	mean|SHAP|	Direction ρ (95% CI)	Expected	Agrees	Ordinal OR (95% CI)
ER positive	0.330	-0.76 (-0.78, -0.74)	↓	yes	0.17 (0.12–0.24)
T category	0.158	+0.82 (+0.81, +0.83)	↑	yes	1.61 (1.31–2.00)
PR positive	0.135	-0.87 (-0.87, -0.86)	↓	yes	0.57 (0.46–0.72)
Node positive	0.086	+0.87 (+0.86, +0.87)	↑	yes	1.29 (1.08–1.62)
HER2 positive	0.065	+0.57 (+0.54, +0.60)	↑	yes	1.83 (1.28–2.69)
Mixed (vs IDC)	0.064	-0.55 (-0.57, -0.52)	–	–	0.53 (0.38–0.74)
Age (per 10 y)	0.061	-0.25 (-0.30, -0.21)	↓	yes	0.93 (0.83–1.03)
ILC (vs IDC)	0.053	-0.47 (-0.49, -0.43)	↓	yes	0.41 (0.28–0.57)
Other type (vs IDC)	0.034	-0.35 (-0.38, -0.31)	–	–	0.26 (0.16–0.43)
Post-menopausal	0.008	-0.65 (-0.68, -0.61)	–	–	0.96 (0.68–1.37)

3.7. Leakage audit

Every shortcut inflated apparent performance (Table 6, Figure 7). Adding the Nottingham Prognostic Index raised the XGBoost macro-AUC from 0.721 to 0.972; adding the Nottingham component scores raised LR from 0.696 to 0.962 in Sanger and from 0.698 to 0.995 in Duke; imputing and oversampling before splitting raised k-NN from 0.652 to 0.806. Splitting by record rather than patient produced a smaller gain (0.544 to 0.579 for 1-NN) because few MSK patients had more than one record. The largest inflation, +0.297 in macro-AUC, exceeds the whole gap between the baseline and the best honest model.

3.8. Imaging extension: DCE-MRI radiomics in Duke

In 637 graded Duke patients, the clinical model reached a cross-validated macro-AUC of 0.694 (0.663–0.727), radiomics alone 0.608 (0.575–0.645) and the

Table 6: Leakage audit: identical models evaluated with the correct and a leaky protocol (5-fold cross-validation).

Scenario	Model	Honest AUC	Leaky AUC	Inflation
Target leakage: NPI as predictor (METABRIC)	Multinomial LR	0.726	0.949	+0.223
Target leakage: NPI as predictor (METABRIC)	XGBoost	0.721	0.972	+0.251
SMOTE + imputation before split (METABRIC)	k-NN	0.652	0.806	+0.154
SMOTE + imputation before split (METABRIC)	XGBoost	0.669	0.850	+0.181
Grade components as predictors (Sanger)	Multinomial LR	0.696	0.962	+0.266
Grade components as predictors (Duke MRI)	Multinomial LR	0.698	0.995	+0.297
Record-level split, 1718 records / 1573 patients (MSK)	k-NN	0.544	0.579	+0.035
Record-level split, 1718 records / 1573 patients (MSK)	Random forest	0.621	0.653	+0.033

fusion model 0.648 (0.618–0.685) (Table 7, Figure 8). The paired gain of fusion over the clinical model was -0.045 (-0.077 to -0.011).

Table 7: Duke-Breast-Cancer-MRI: clinical, radiomics and fusion models (nested 10×5-fold cross-validation; L_2-penalised multinomial LR; paired bootstrap Δ versus the clinical model).

Model (L2 logistic)	Features	Macro-AUC (95% CI)	G3-AUC (95% CI)	Δ macro-AUC vs clinical (95% CI)	Brier
Clinical (full set)	10	0.694 (0.663–0.727)	0.769 (0.731–0.811)	ref.	0.551
DCE-MRI radiomics (529 features)	529	0.608 (0.575–0.645)	0.667 (0.626–0.718)	-0.085 (-0.126, -0.045)	0.600
Clinical + radiomics	539	0.648 (0.618–0.685)	0.731 (0.693–0.777)	-0.045 (-0.077, -0.011)	0.582

3.9. Transport to Saudi case-mix

Raking METABRIC to the Dammam receptor, histology and age profile (ESS 1088 of 1861) moved its grade distribution to 6.2/32.9/60.9% (grade 3 95% CI 58.5–63.0%) (Table 8, Figure 9), compared with 50.6% observed and 36.7% published. The same direction held for MSK-2018 (77.3%) and Duke (42.1%), for the assumed share aged ≤50 years between 40% and 60% (grade 3 60.5–61.3%; Supplementary Table S7), and for the Riyadh anchor (METABRIC re-weighted grade 3 57.3%, 54.6–60.0%, against 48.8% published). Across 3 Saudi series (908 patients) the pooled published grade-3 share was 44.7% (37.2–52.5%). Under Dammam case-mix, LR discrimination rose from 0.727 to 0.790 in METABRIC, and the mean predicted probability of grade 3 was 59.1%, about 22 percentage points above the published Dammam figure.

Table 8: Re-weighting to published Saudi case-mix. ESS = effective sample size of the raking weights (first row of each block: patients with complete raking variables). Grade-3 CI: bootstrap with re-raking. Arrows: unweighted→Saudi-weighted LR discrimination. †Dammam series [27]; ‡Riyadh series [28] (tertiary referral centre). Pooled: DerSimonian–Laird random effects across 3 Saudi series reporting grade [27, 28, 64]; the Madinah series reports grade for invasive ductal carcinoma only and no receptor status, so it contributes to the pooled estimate but not to raking.

Anchor	Cohort	Quantity	G1	G2	G3 (95% CI)	ESS	Macro-AUC	G3-AUC
Dammam (Kussaibi 2021)†	METABRIC	observed	8.9	40.5	50.6	1861		
		re-weighted	6.2	32.9	60.9 (58.5–63.0)	1088		
		LR predicted (weighted)	6.7	34.2	59.1		0.727→0.790	0.759→0.811
		published	8.7	54.5	36.7			
Dammam (Kussaibi 2021)†	MSK-2018	observed	5.7	26.7	67.6	1496		
		re-weighted	3.4	19.2	77.3 (75.1–79.9)	846		
		LR predicted (weighted)	5.6	33.1	61.3		0.715→0.766	0.734→0.765
		published	8.7	54.5	36.7			
Dammam (Kussaibi 2021)†	Duke MRI	observed	17.7	49.8	32.5	637		
		re-weighted	13.2	44.7	42.1 (37.7–45.9)	510		
		LR predicted (weighted)	5.8	32.8	61.4		0.709→0.717	0.779→0.755
		published	8.7	54.5	36.7			
Riyadh (Durrani 2025)‡	METABRIC	observed	8.9	40.5	50.6	1861		
		re-weighted	8.2	34.5	57.3 (54.6–60.0)	852		
		LR predicted (weighted)	8.7	35.6	55.7		0.727→0.785	0.759→0.806
		published	4.2	47.0	48.8			
Riyadh (Durrani 2025)‡	MSK-2018	observed	5.7	26.7	67.6	1496		
		re-weighted	4.3	19.5	76.3 (73.8–78.6)	555		
		LR predicted (weighted)	6.8	34.8	58.4		0.715→0.775	0.734→0.771
		published	4.2	47.0	48.8			
Riyadh (Durrani 2025)‡	Duke MRI	observed	17.7	49.8	32.5	637		
		re-weighted	14.0	50.1	35.9 (32.5–39.7)	277		
		LR predicted (weighted)	8.3	34.5	57.1		0.709→0.710	0.779→0.776
		published	4.2	47.0	48.8			
Pooled Saudi anchors (random effects)		published	9.2	–	44.7 (37.2–52.5)			

3.10. Sensitivity analysis

In the Ljubljana cohort (286 patients, binned covariates, no receptor data) the best macro-AUC for "degree of malignancy" was 0.636 (Supplementary Table S8), consistent with the loss of discrimination whenever receptor status is unavailable.

4. Discussion

4.1. Principal findings

Using only open data, routinely recorded clinicopathological variables carried moderate and transportable information about Nottingham grade: macro-AUCs of 0.72–0.73 internally and 0.64–0.80 externally, with grade 3 the most and grade 2 the least predictable class. Hormone-receptor status dominated both the univariable associations and the model explanations, and tumour size, nodal status, HER2 and lobular histology contributed in clinically expected directions. Model flexibility did not help: in paired comparisons no flexible learner improved on multinomial logistic regression internally, and only isolated external comparisons differed in either direction, in line with systematic comparisons of clinical prediction models [17] and with the small number of mostly binary predictors available. Discrimination and predictor effects transported across countries, eras and a cohort recruited from MRI rather than genomics (calibration slopes close to one), but calibration-in-the-large did not: the same frozen model under-predicted grade 3 in MSK-2018 and over-predicted it in Duke, because the baseline frequency of grade 3 differs between sources far more than their predictors do. Any use outside the development population therefore needs, at minimum, a local intercept update, which fully corrected the miscalibration in both external cohorts.

4.2. What the leakage audit and the imaging extension add

Each common analytic shortcut inflated apparent discrimination, by up to +0.297 in macro-AUC—more than the whole distance between a useless and the best honest model. Component scores and the Nottingham Prognostic Index produce near-perfect "prediction" because they are grade; preprocessing before the split produces smaller but still material gains. This is a practical warning for the grade-prediction literature, where public datasets routinely ship these columns [14, 16]. The Duke extension was equally sobering: 529 DCE-MRI radiomic features did not improve on the routine record, and adding them to it lowered cross-validated discrimination (paired Δ -0.045, -0.077 to -0.011). With 637 graded patients and hundreds of correlated features, the penalised model could not extract signal beyond what receptor status, size and nodes already supply. Reported radiomics AUCs for grade in single-centre studies [12, 13] should be read against a clinical baseline of this kind, evaluated in the same patients.

4.3. What this means for Saudi Arabia

Re-weighting to Saudi case-mix moved the grade-3 share of every source cohort upwards by +9.6 to +10.3 percentage points: the Saudi profile—fewer receptor-positive and more HER2-positive tumours—is, by itself, associated with more high-grade disease. The level reached, however, depended on the source cohort (42.1–77.3% for the Dammam anchor), because the observed grade-3 share already differed between the Western cohorts (32.5–67.6%). Published Saudi series report less grade 3 than the METABRIC- and MSK-based projections (36.7% in Dammam; 48.8% in Riyadh; pooled 44.7% across 3 series), and a METABRIC-developed model would predict on average 59.1% grade 3 in Dammam-like patients, about 22 percentage points too many. Grading practice (grade 2 is the least reproducible category [4, 6]), referral and era effects in hospital series, and genuine differences in tumour biology could all contribute, and aggregate data cannot separate them. Two practical conclusions follow. First, a Western grade model should not be applied to Saudi patients without local recalibration of its intercept, which requires only the local grade distribution. Second, the most valuable next step is a de-identified, grade-labelled Saudi cohort released under open terms; the code here accepts one as an additional harmonisation function and would immediately provide its external validation and recalibration.

4.4. Strengths and limitations

The strengths are full reproducibility from public data with recorded checksums, a pre-specified analysis, five external cohorts from three continents (one recruited through imaging rather than genomics), paired model comparisons, and explicit quantification of leakage. The limitations are substantial. Grade was assigned by different pathologists in different eras under slightly different conventions, and the Ljubljana "degree of malignancy" predates the Elston–Ellis modification; inter-observer variability [4, 5] caps the attainable discrimination. Several cohorts were assembled for genomic studies or enriched for advanced disease and are not population-based. Tumour size was harmonised to T category, and T and N were unavailable in three external cohorts. The Riley criteria were not met: the grade 1 versus grade 2 sub-model, whose anticipated R^2 is very small, would require 4477 patients for the conventional shrinkage target, against 2386 available. Penalised estimation, internal calibration slopes close to one and the stable external slopes suggest that overfitting was limited in practice, but estimates for the grade 1/2 boundary should be read with this in mind. The Saudi analysis rests on published marginal distributions rather than patient data; the Dammam series does not report the share aged $\leq$50 years (results were insensitive to the assumed value); the Riyadh series comes from a tertiary referral centre with many stage IV patients and includes about 10% non-Saudi patients; and the Madinah series reports grade for ductal

carcinoma only. Published series also differ from one another in era, region and referral pattern, and their pooled grade-3 share is correspondingly heterogeneous. The pre-specified tabular foundation model (TabPFN) could not be run because its weights are distributed through a gated repository that was not reachable from the build environment; the code path is in place (`make tabpfn`). The image-level arm (Saudi KAU-BCMD mammograms [65], Egyptian CDD-CESM and BUSI collections [66, 67]) was not run for the same reason.

4.5. Implications and next steps

For research, a routine-variable model of this kind is the minimum baseline that image-based grade predictors should beat in the same patients, under patient-level splitting and with the leakage columns removed. For practice, such a model is not a substitute for histological grading; its plausible uses are imputing missing grade in registry or trial data with calibrated uncertainty, and flagging records whose assigned grade is improbable given the rest of the record for review. Any such use requires local recalibration. The repository makes it straightforward to add a Saudi or Gulf cohort, the TabPFN rung and the image-level arm as soon as the data become accessible.

5. Conclusions

Routine clinicopathological variables predict breast cancer grade moderately and consistently across open cohorts from several countries. Logistic regression is sufficient, MRI radiomics added nothing in the cohort where they could be tested, external calibration needs a local intercept update, and leaky protocols can manufacture apparently excellent models. For Saudi Arabia, case-mix alone implies more high-grade disease than in Western cohorts, yet Western-trained models would over-predict grade 3 relative to published Saudi series, which makes an openly shared, grade-labelled Saudi cohort the highest-value next contribution.

Data and code availability

All data are publicly available from their custodians: cBioPortal studies `brca_metabric`, `breast_msk_2018`, `brca_mbcproject_2022`, `brca_sanger`, `brca_broad`, `brca_bccrc` and `brca_tcga_pan_can_atlas_2018`; the TCIA Duke-Breast-Cancer-MRI clinical and imaging-feature tables (CC BY-NC 4.0; non-commercial use); and the UCI Breast Cancer dataset (doi:10.24432/C51P4M). Data are not redistributed: a download script retrieves them and verifies the SHA-256 checksums recorded in the repository (files retrieved 17–18 October 2026). Code, configuration, result files and the manuscript sources are available

at https://anonymous.4open.science/r/bc-grade-opendata; `make all` reproduces every table, figure and in-text number.

Ethics statement

Secondary analysis of de-identified, publicly released data; no ethics approval or consent was required. The study was not registered.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude (Anthropic) to assist with analysis code, reference checking and drafting of the text. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. [Confirm or edit to reflect actual use.]

CRediT authorship contribution statement

Aminat Abiola Ajibola: Conceptualization, Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. [Add co-authors and their roles if any.]

Declaration of competing interest

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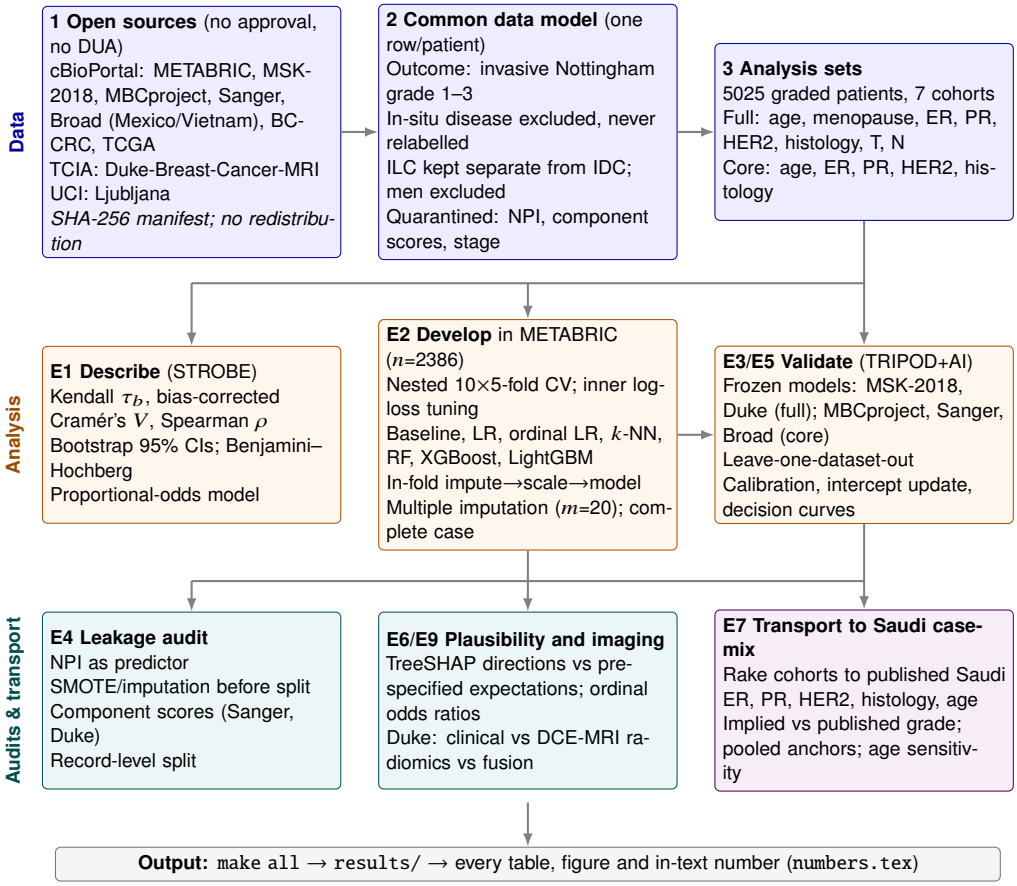


Figure 1: Study framework. **A**, open data are downloaded with checksums and mapped to one row per patient under four harmonisation rules; variables that encode grade are quarantined. **B**, associations are described with scale-appropriate statistics; models are developed in METABRIC with nested cross-validation and validated, frozen, in five external cohorts and by leave-one-dataset-out. **C**, the results are audited for leakage and clinical plausibility, extended with MRI radiomics in the Duke cohort, and transported to published Saudi case-mix. E1–E10 are the experiment stages in the repository.

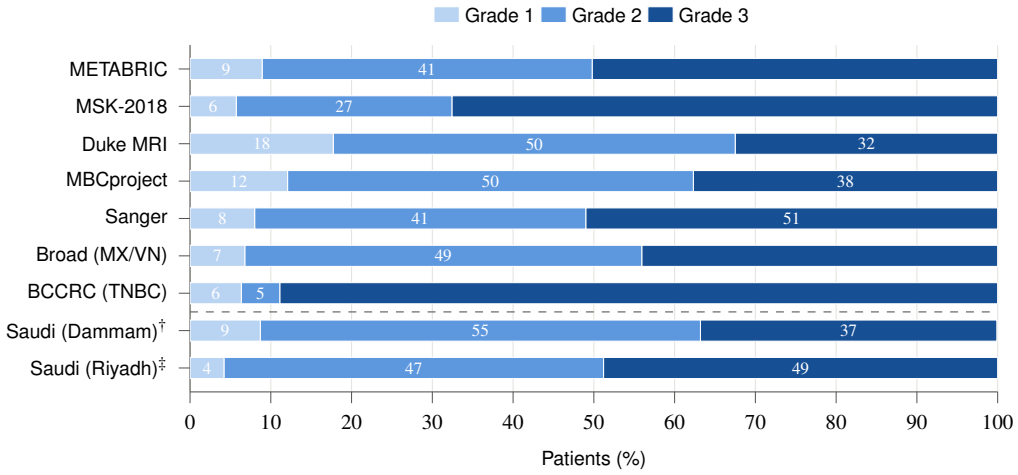


Figure 2: Distribution of Nottingham grade by cohort (%), with the published Saudi series below the dashed line (†27; ‡28).

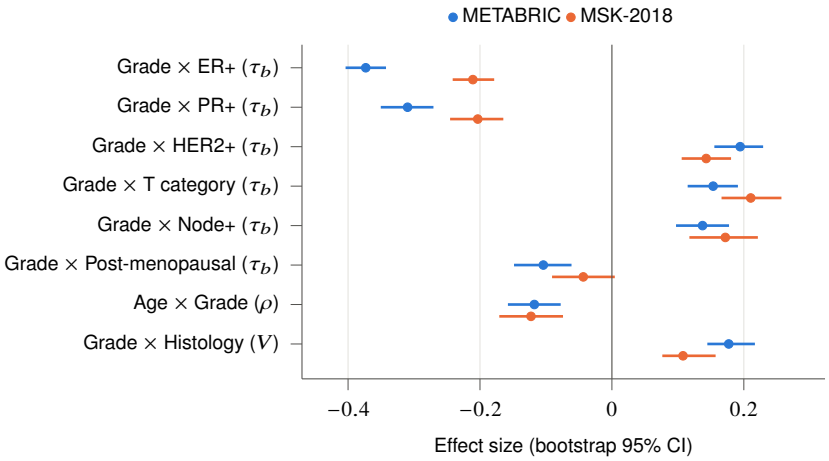


Figure 3: Associations involving grade in the two large cohorts (effect and bootstrap 95% CI).

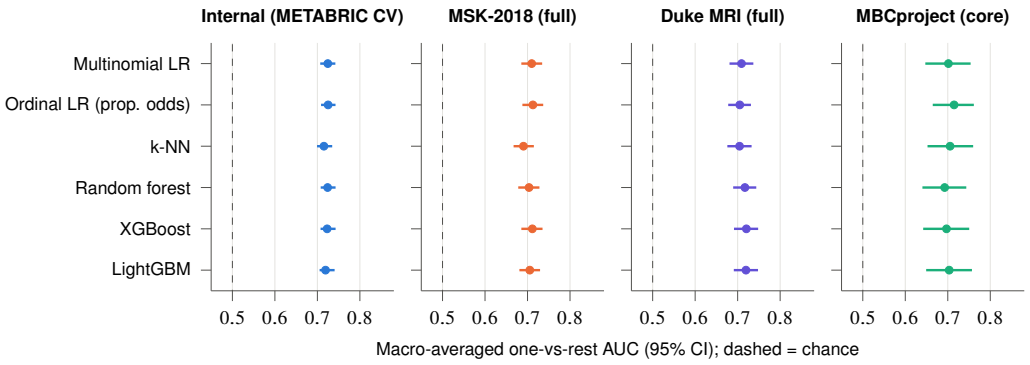


Figure 4: Macro-averaged AUC (95% CI) of the model ladder internally and in three external cohorts; dashed line = chance.

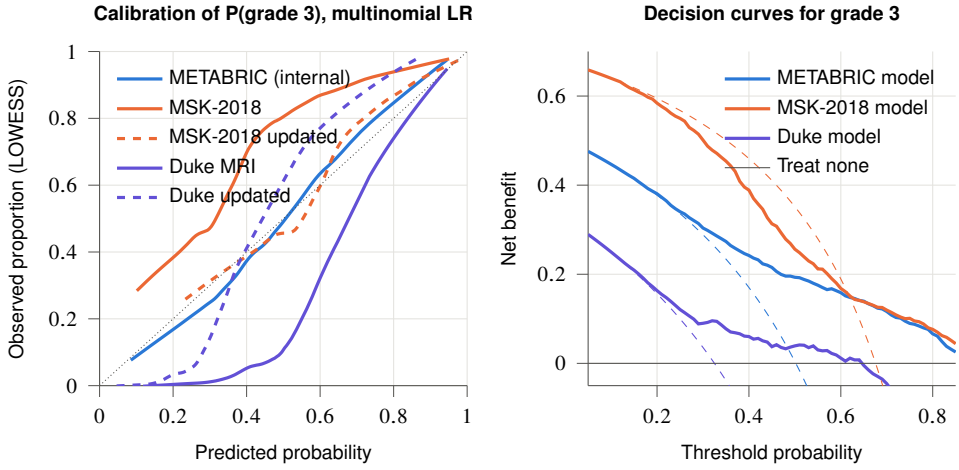


Figure 5: Left: LOWESS calibration curves for P(grade 3) from multinomial LR, internally, externally (solid) and after a cross-fitted intercept update (dashed). Right: decision curves (solid = model; thin dashed = treat all, same colour; treat none = 0).

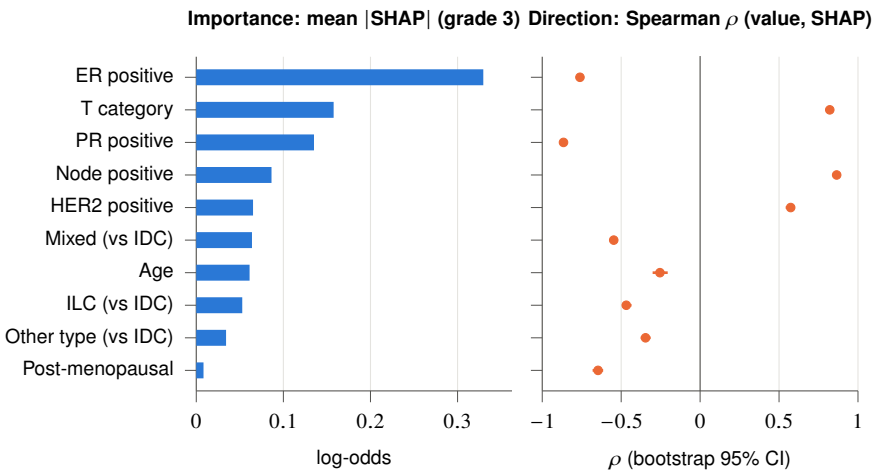


Figure 6: Global importance (mean |SHAP|) and direction (Spearman ρ between feature value and SHAP value, bootstrap 95% CI) for the grade-3 output of XGBoost in METABRIC. Beeswarm in Supplementary Figure S1.

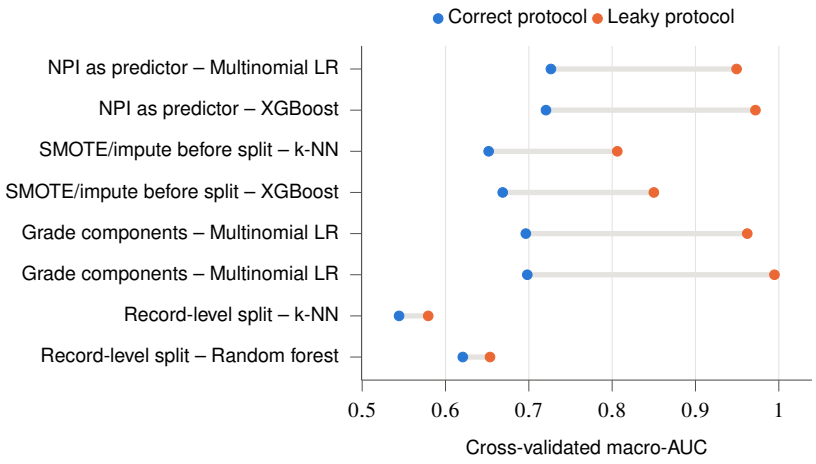


Figure 7: Apparent macro-AUC under the correct (blue) and leaky (orange) protocols.

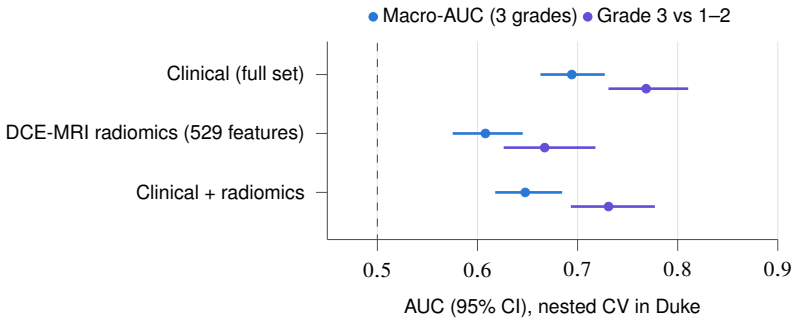


Figure 8: Duke: discrimination of clinical, radiomics and fusion models (95% CI); dashed = chance.

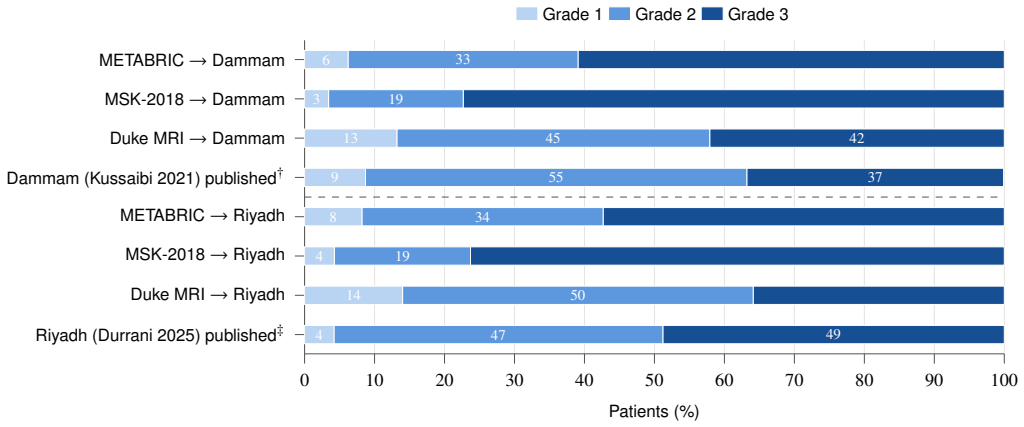


Figure 9: Grade distributions implied by re-weighting the development and external cohorts to each Saudi anchor's case-mix, and the published Saudi distributions.

Supplementary material

Table S1: Participant flow (STROBE).

Cohort	Step	*n*
METABRIC	records	2509
METABRIC	invasive carcinoma, female (excl. stage 0 / sarcoma)	2484
METABRIC	analysis set (grade 1-3 recorded)	2386
MSK-2018	samples	1918
MSK-2018	patients	1756
MSK-2018	female	1746
MSK-2018	analysis set (grade 1-3 recorded)	1573
MBCproject	patients	301
MBCproject	female, abstraction complete	256
MBCproject	invasive carcinoma	254
MBCproject	analysis set (grade 1-3 recorded)	207
Sanger	patients	100
Sanger	analysis set (grade 1-3 recorded)	100
Broad (MX/VN)	patients	103
Broad (MX/VN)	invasive carcinoma	94
Broad (MX/VN)	analysis set (grade 1-3 recorded)	59
BCCRC (TNBC)	patients	65
BCCRC (TNBC)	carcinoma	64
BCCRC (TNBC)	analysis set (grade 1-3 recorded)	63
Duke MRI	patients	922
Duke MRI	invasive carcinoma	921
Duke MRI	analysis set (grade 1-3 recorded)	637
TCGA-BRCA	patients	1084
TCGA-BRCA	female	1072

Table S2: Leave-one-dataset-out validation (core feature set).

Held-out cohort	Model	n	Macro-AUC (95% CI)	G3-AUC (95% CI)	Macro-F1
METABRIC	Multinomial LR	2386	0.698 (0.679–0.716)	0.727 (0.706–0.747)	0.393
METABRIC	Ordinal LR (prop. odds)	2386	0.696 (0.678–0.713)	0.728 (0.707–0.748)	0.413
METABRIC	Random forest	2386	0.702 (0.683–0.720)	0.730 (0.709–0.749)	0.410
METABRIC	XGBoost	2386	0.700 (0.681–0.718)	0.725 (0.704–0.745)	0.412
METABRIC	LightGBM	2386	0.689 (0.671–0.706)	0.724 (0.704–0.745)	0.433
MSK-2018	Multinomial LR	1573	0.659 (0.632–0.685)	0.684 (0.657–0.711)	0.285
MSK-2018	Ordinal LR (prop. odds)	1573	0.670 (0.645–0.695)	0.681 (0.654–0.709)	0.257
MSK-2018	Random forest	1573	0.664 (0.638–0.691)	0.676 (0.649–0.703)	0.302
MSK-2018	XGBoost	1573	0.662 (0.635–0.689)	0.672 (0.645–0.700)	0.315
MSK-2018	LightGBM	1573	0.661 (0.634–0.688)	0.673 (0.645–0.700)	0.335
MBCproject	Multinomial LR	207	0.709 (0.656–0.759)	0.778 (0.713–0.840)	0.322
MBCproject	Ordinal LR (prop. odds)	207	0.716 (0.665–0.763)	0.771 (0.706–0.835)	0.483
MBCproject	Random forest	207	0.707 (0.658–0.756)	0.770 (0.703–0.835)	0.332
MBCproject	XGBoost	207	0.710 (0.658–0.760)	0.773 (0.705–0.839)	0.358
MBCproject	LightGBM	207	0.714 (0.664–0.764)	0.764 (0.697–0.831)	0.329
Sanger	Multinomial LR	100	0.778 (0.692–0.855)	0.787 (0.692–0.871)	0.483
Sanger	Ordinal LR (prop. odds)	100	0.759 (0.675–0.838)	0.791 (0.696–0.871)	0.464
Sanger	Random forest	100	0.769 (0.683–0.847)	0.786 (0.690–0.869)	0.391
Sanger	XGBoost	100	0.774 (0.683–0.857)	0.791 (0.695–0.875)	0.442
Sanger	LightGBM	100	0.761 (0.670–0.845)	0.791 (0.695–0.874)	0.455
Duke MRI	Multinomial LR	637	0.703 (0.676–0.730)	0.782 (0.743–0.820)	0.459
Duke MRI	Ordinal LR (prop. odds)	637	0.705 (0.678–0.732)	0.782 (0.744–0.820)	0.465
Duke MRI	Random forest	637	0.717 (0.688–0.744)	0.787 (0.749–0.825)	0.460
Duke MRI	XGBoost	637	0.712 (0.682–0.739)	0.782 (0.743–0.822)	0.446
Duke MRI	LightGBM	637	0.708 (0.678–0.736)	0.780 (0.741–0.819)	0.458

Table S3: Calibration of P(grade 3): intercept (calibration-in-the-large) and slope, internally, in the external full-feature cohorts and after a cross-fitted intercept update.

Model	Setting	n	Obs. G3	Mean pred.	Intercept	Slope	G3-AUC
Multinomial LR	METABRIC (internal CV)	2386	0.502	0.502	-0.00	1.00	0.753
Multinomial LR	MSK-2018 (external)	1573	0.676	0.472	+0.99	1.03	0.727
Multinomial LR	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.03	0.727
Multinomial LR	Duke MRI (external)	637	0.325	0.517	-1.02	0.96	0.779
Multinomial LR	Duke MRI (intercept-updated)	637	0.325	0.325	+0.00	0.96	0.779
Ordinal LR (prop. odds)	METABRIC (internal CV)	2386	0.502	0.501	+0.00	0.96	0.752
Ordinal LR (prop. odds)	MSK-2018 (external)	1573	0.676	0.474	+0.97	1.02	0.725
Ordinal LR (prop. odds)	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.02	0.724
Ordinal LR (prop. odds)	Duke MRI (external)	637	0.325	0.520	-1.05	0.92	0.778
Ordinal LR (prop. odds)	Duke MRI (intercept-updated)	637	0.325	0.325	+0.00	0.92	0.777
XGBoost	METABRIC (internal CV)	2386	0.502	0.503	-0.00	1.02	0.755
XGBoost	MSK-2018 (external)	1573	0.676	0.474	+0.98	1.03	0.726
XGBoost	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.03	0.726
XGBoost	Duke MRI (external)	637	0.325	0.519	-1.06	0.97	0.789
XGBoost	Duke MRI (intercept-updated)	637	0.325	0.325	-0.00	0.97	0.789

Table S4: Multiclass calibration-in-the-large: observed versus mean predicted prevalence of each grade (%).

Model	Setting	Obs. G1	Pred. G1	Obs. G2	Pred. G2	Obs. G3	Pred. G3
Multinomial LR	METABRIC (internal CV)	8.9	8.9	40.9	40.9	50.2	50.2
Multinomial LR	MSK-2018 (external)	5.7	8.5	26.7	44.3	67.6	47.2
Multinomial LR	Duke MRI (external)	17.7	7.6	49.8	40.7	32.5	51.7
Ordinal LR (prop. odds)	METABRIC (internal CV)	8.9	9.0	40.9	40.9	50.2	50.1
Ordinal LR (prop. odds)	MSK-2018 (external)	5.7	9.1	26.7	43.5	67.6	47.4
Ordinal LR (prop. odds)	Duke MRI (external)	17.7	8.1	49.8	39.9	32.5	52.0
XGBoost	METABRIC (internal CV)	8.9	8.9	40.9	40.8	50.2	50.3
XGBoost	MSK-2018 (external)	5.7	8.7	26.7	43.8	67.6	47.4
XGBoost	Duke MRI (external)	17.7	8.2	49.8	39.9	32.5	51.9

Table S5: Missing data: multiple imputation with pooled predictions and complete-case analysis versus the primary single in-fold imputation (METABRIC, nested cross-validation).

Model	Missing-data method	n	Macro-AUC (95% CI)	G3-AUC	Macro-F1	Brier
Multinomial LR	Multiple imputation (m=20, pooled predictions)	2386	0.724 (0.707–0.742)	0.752	0.465	0.490
Multinomial LR	Complete-case analysis	1800	0.729 (0.709–0.747)	0.761	0.455	0.486
Multinomial LR	Single in-fold imputation (primary)	2386	0.725 (0.707–0.742)	0.752	0.465	0.490
XGBoost	Multiple imputation (m=20, pooled predictions)	2386	0.726 (0.710–0.745)	0.755	0.465	0.489
XGBoost	Complete-case analysis	1800	0.727 (0.709–0.747)	0.760	0.453	0.487
XGBoost	Single in-fold imputation (primary)	2386	0.723 (0.708–0.743)	0.753	0.458	0.492

Table S6: Minimum sample size for developing the multinomial model (Riley criteria applied to each pairwise sub-model).

Sub-model	Share of patients	Anticipated R^2_{CS}	Max R^2_{CS}	n (criterion i)	n (criterion ii)
G1 vs G2	0.498	0.039	0.610	4477	636
G1 vs G3	0.591	0.180	0.572	758	528
G2 vs G3	0.911	0.171	0.747	521	263
Criterion (iii): each grade prevalence ±0.05					385
Required (maximum) / available					4477 / 2386

Table S7: Sensitivity of the Saudi-weighted METABRIC grade distribution (%) to the assumed share of patients aged ≤50 years in the Dammam series (which reports only the median age).

Assumed share ≤50 y	ESS	G1	G2	G3
40%	1278	6.3	33.2	60.5
45%	1187	6.3	33.0	60.7
50%	1088	6.2	32.9	60.9
55%	988	6.2	32.7	61.1
60%	892	6.1	32.6	61.3

Table S8: Ljubljana sensitivity analysis ("degree of malignancy"; age band, menopausal status, tumour-size band, nodes, node capsule invasion; 10×5-fold CV).

Model	n	Macro-AUC (SD over repeats)	Macro-F1	G3-AUC
Majority class	286	0.497 (0.000)	0.208	0.498
Multinomial LR	286	0.636 (0.011)	0.425	0.691
Ordinal LR (prop. odds)	286	0.618 (0.010)	0.397	0.694
Random forest	286	0.626 (0.018)	0.392	0.698
XGBoost	286	0.622 (0.014)	0.402	0.686

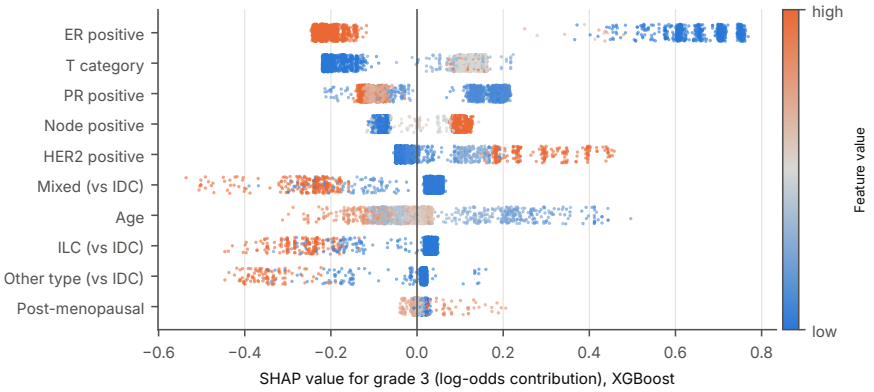


Figure S1: SHAP beeswarm for the grade-3 output of XGBoost in METABRIC; colour encodes the (imputed) feature value.

Table S9: TRIPOD+AI checklist: where each item is reported.

Item	Topic	D/E	Location in this paper
1	Title	D;E	Title: prediction of grade, development and external validation, open data
2	Abstract	D;E	Structured abstract (TRIPOD+AI for Abstracts items)
3a	Background: healthcare context	D;E	Introduction, paragraphs 1 and 3
3b	Background: target population and intended use	D;E	Introduction, paragraph 1 (baseline for image models, registry imputation, quality assurance); Discussion, *Implications*
3c	Background: known biases/inequalities	D;E	Introduction, paragraph 3 (Saudi/Gulf case-mix, absence of open regional data)

Item	Topic	D/E	Location
4	Objectives	D;E	Introduction, *Contributions*
5a	Data sources	D;E	Methods, *Data sources*; Table 1
5b	Dates of data collection	D;E	Methods, *Data sources*; Supplementary Table S1; Data availability (download date and checksums)
6a	Setting and centres	D;E	Methods, *Data sources*
6b	Eligibility criteria	D;E	Methods, *Harmonisation* (rules i–iv)
6c	Treatments received	D;E	Not applicable at diagnosis (grade is a baseline characteristic); Limitations
7	Data preparation	D;E	Methods, *Harmonisation*; repository `harmonise.py`
8a	Outcome definition	D;E	Methods, *Outcome*: invasive Nottingham grade 1–3; grade 3 vs 1–2
8b	Outcome assessment	D;E	Assigned by pathologists at each source; differences discussed under Limitations
8c	Blinding of outcome assessment	D;E	Not applicable (retrospective records); Limitations
9a	Choice of predictors	D	Methods, *Outcome, predictors and feature sets*
9b	Predictor definitions	D;E	Methods, *Harmonisation*; repository code
9c	Blinding of predictor assessment	D;E	Not applicable (routine records); leakage quarantine
10	Sample size	D;E	Methods, *Sample size*; Supplementary Table S6
11	Missing data	D;E	Methods, *Missing data*; Supplementary Table S5
12a	Data use (development/evaluation)	D;E	Methods, *Validation*; Figure 1
12b	Pre-processing	D	Methods, *Model development* (in-fold imputation and scaling)
12c	Model type and building	D	Methods, *Model development*
12d	Heterogeneity across clusters	D;E	LODO validation; cohort-level results (Tables 4, S2)
12e	Performance measures	D;E	Methods, *Performance measures*
12f	Model updating	E	Methods, *Performance measures* (intercept update)
12g	Comparison with other models	D;E	Model ladder; paired Δ vs LR (Table 3)
13	Class imbalance	D;E	Methods, *Model development* (no resampling; prevalence-adjusted rule)

Item	Topic	D/E	Location
14	Fairness	D;E	Results by country/cohort; Saudi transport (Section 2.12)
15	Model output	D	Probabilities for each grade
16	Training versus evaluation differences	D;E	Methods, *Validation*; frozen external models
17	Ethical approval	D;E	Declarations
18a	Funding	D;E	Declarations
18b	Conflicts of interest	D;E	Declarations
18c	Protocol	D;E	Pre-specified analysis plan in `configs/experiment.yaml` (repository)
18d	Registration	D;E	Not registered (secondary analysis of open data); Declarations
18e	Data sharing	D;E	Data availability
18f	Code sharing	D;E	Code availability
19	Patient and public involvement	D;E	Declarations
20a	Participant flow	D;E	Supplementary Table S1
20b	Participant characteristics	D;E	Table 1; Figure 2
20c	Comparison of development and evaluation data	E	Table 1; Results, *Cohorts*
21	Number of participants and outcome events	D;E	Tables 1, 3, 4
22	Full model specification	D	Code and fitted coefficients in the repository; ordinal odds ratios (Table 5)
23a	Performance with CIs	D;E	Tables 3, 4; Figure 4
23b	Performance in subgroups	D;E	By cohort/country; Saudi-weighted performance (Table 8)
24	Model updating results	E	Figure 5; Supplementary Table S3
25	Interpretation	D;E	Discussion, *Principal findings*
26	Limitations	D;E	Discussion, *Strengths and limitations*
27a	Usability: handling poor-quality or missing input	D	Methods, *Missing data*; Discussion

Item	Topic	D/E	Location
27b	Usability: user interaction and expertise	D	Discussion, *Implications*
27c	Next steps and generalisability	D;E	Discussion, *Implications and next steps*